

Project Synopsis

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Date of Submission	15-08-2026

Title

Analyzing Operational Efficiency and Predicting Breakdown Risk and Shipment Weight in the Warehouse Network of Supply Flow FMCG Solutions

Executive Summary

Supply Flow FMCG Solutions distributes instant noodles through a warehouse network where performance varies widely. Some warehouses meet demand on time, while others face repeated storage, transport and staffing problems, and breakdowns halt work without warning. The project covers three areas:

- Identifying the conditions that separate strong warehouses from weak ones.
- Estimating breakdown risk so that preventive action can be planned.
- Recommending a suitable shipment weight for each warehouse.

It sits within supply chain analytics, applying predictive machine learning to the records the company already maintains, to support decisions on infrastructure spending, maintenance planning and dispatch quantity.

Objectives and Scope

Objectives

1. To find which operational and infrastructure conditions explain the differences in shipment weight and reported problems across the 25,000 warehouses.
2. To build a model that estimates whether a warehouse will report a breakdown.
3. To predict a suitable shipment weight for each warehouse and compare volumes across the four zones and six regional zones.
4. To provide a unified solution enabling insights into warehouse operational efficiency, warehouse breakdown risk and per warehouse suitable shipment weight powered by predictive machine learning, rather than just depending on descriptive dashboards and heuristic models.

Scope

- All 25,000 warehouses in the company file, spread across four zones and six regional zones, in rural and urban locations, and both companies owned and rented.
- Offers a descriptive summary of the network, followed by a data mining analysis of the conditions linked to differences in operational efficiency.
- Delivers predictive models that estimate breakdown risk and a suitable shipment weight for each warehouse, with the factors that matter most in each.
- Excludes prescriptive recommendations and optimization frameworks. The study explains and predicts, but does not compute optimal allocations, routes, costs or pricing.

Methodology

Type of Research

- Exploratory: identifies the conditions linked to differences in warehouse performance.
- Empirical: delivers conclusions and models tested against recorded data.

Data Collection

- Mode: Secondary data collection.
- Source: data collected by Supply Flow FMCG Solutions, comprising 25,000 rows and 24 columns, with each row corresponding to an individual warehouse and its attributes.

Data Preparation

- Data type check: confirming each field is stored in the correct format.
- Duplication check: identifying and removing duplicate records.
- Missing value treatment: removal or imputation, depending on the extent of the gap.
- Outlier analysis: IQR and other statistical methods to identify extreme values and decide whether to include or exclude them.
- Feature engineering: creating derived variables from existing fields and dropping those that add no explanatory value.
- Qualitative data treatment: label encoding, ordinal encoding and one hot encoding.
- Feature scaling: applied where the algorithm is sensitive to differences in range.

- Data imbalance check: upsampling or downsampling of the training data, applied as required by each objective.

Exploratory Data Analysis

- Univariate, bivariate and multivariate analysis using visualization tools.
- Correlation analysis to understand pairwise relationships between variables.
- Hypothesis tests such as chi square and analysis of variance to confirm association between features.

Model Building

- Train and test split, with the test set held back for final assessment.
- Clustering algorithms such as K-Means and hierarchical clustering to form performance segments, followed by cluster profiling to identify the drivers within each segment.
- Classification algorithms for breakdown risk: logistic regression as the baseline, followed by Naive Bayes, support vector machines, decision trees, random forest, XGBoost, AdaBoost and CatBoost.
- Regression algorithms for shipment weight: linear regression as the baseline, followed by support vector regression, decision trees, random forest, XGBoost, AdaBoost and CatBoost.
- K-fold cross validation and hyperparameter tuning to address the bias variance trade off.

Evaluation

- Clustering: silhouette score, Dunn index and cluster profiling.
- Classification: accuracy, precision, recall, F1 score, ROC-AUC and PR-AUC.
- Regression: R squared, adjusted R squared, MAE, MSE, RMSE and MAPE.

Tools

- Python as the working language, with VS Code and JupyterLab as the development environment and uv for package management.
- pandas and NumPy for data handling, scikit-learn, XGBoost and PyCaret for modelling, and Optuna for hyperparameter tuning.
- Matplotlib, Seaborn and Plotly for visualization, and Streamlit for presenting the results.

Preliminary Findings and Expected Results

Warehouse performance is currently managed through two kinds of tools:

- Descriptive dashboards that report refills, storage issues, breakdowns and shipment volumes by warehouse and zone, showing what has already happened.
- Heuristic rules such as fixed reorder points and threshold alerts, set by judgement and applied uniformly across the network.

Both react after a problem appears, weigh one measure at a time, and cannot be tested for accuracy. Standard predictive methods do not fill the gap either, since time series forecasting needs repeated observations and predictive maintenance needs sensor readings, neither of which this data holds.

A predictive model built on the recorded warehouse conditions offers what the current tools cannot. It weighs all conditions together, gives an estimate for each warehouse before the event, and is validated on data held back from training, so its error is known.

Initial checks suggest shipment weight is driven mainly by reported storage issues and warehouse age, while location and ownership appear weak. The models are expected to rank these factors and flag the warehouses most at risk. Since the data is a single snapshot, the results show association rather than cause.